

Occurrence of pharmaceuticals and personal care products in fish: results of a national pilot study in the United States.

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Pharmaceuticals and personal care products are being increasingly reported in a variety of biological matrices, including fish tissue; however, screening studies have presently not encompassed broad geographical areas. A national pilot study was initiated in the United States to assess the accumulation of pharmaceuticals and personal care products in fish sampled from five effluent-dominated rivers that receive direct discharge from wastewater treatment facilities in Chicago, Illinois; Dallas, Texas; Orlando, Florida; Phoenix, Arizona; and West Chester, Pennsylvania, USA. Fish were also collected from the Gila River, New Mexico, USA, as a reference condition expected to be minimally impacted by anthropogenic influence. High performance liquid chromatography-tandem mass spectrometry analysis of pharmaceuticals revealed the presence of norfluoxetine, sertraline, diphenhydramine, diltiazem, and carbamazepine at nanogram-per-gram concentrations in fillet composites from effluent-dominated sampling locations; the additional presence of fluoxetine and gemfibrozil was confirmed in liver tissue. Sertraline was detected at concentrations as high as 19 and 545 ng/g in fillet and liver tissue, respectively. Gas chromatography-tandem mass spectrometry analysis of personal care products in fillet composites revealed the presence of galaxolide and tonalide at maximum concentrations of 2,100 and 290 ng/g, respectively, and trace levels of triclosan. In general, more pharmaceuticals were detected at higher concentrations and with greater frequency in liver than in fillet tissues. Higher lipid content in liver tissue could not account for this discrepancy as no significant positive correlations were found between accumulated pharmaceutical concentrations and lipid content for either tissue type from any sampling site. In contrast, accumulation of the personal care products galaxolide and tonalide was significantly related to lipid content. Results suggest that the detection of pharmaceuticals and personal care products was dependent on the degree of wastewater treatment employed.